

Task 2 Supplementary Architecture Notes

Distributed EV Charging Navigation System

This document supplements the presentation by enlarging the architecture diagrams and recording the assumptions needed to interpret them. The design models 64 charging nodes and four base stations on a cluster. MPI carries inter-process messages, while OpenMP accelerates each base station's local candidate search.

Logical topology	8 by 8 non-periodic mesh with direct logical-neighbour communication
Parallel model	64 single-threaded node ranks; four base ranks with four OpenMP threads each
Deployment	Four 32-core hosts; 20 active cores per host; 80 active cores overall
Communication	Point-to-point operational protocol; Bcast for setup and Barrier for round completion

Design scope

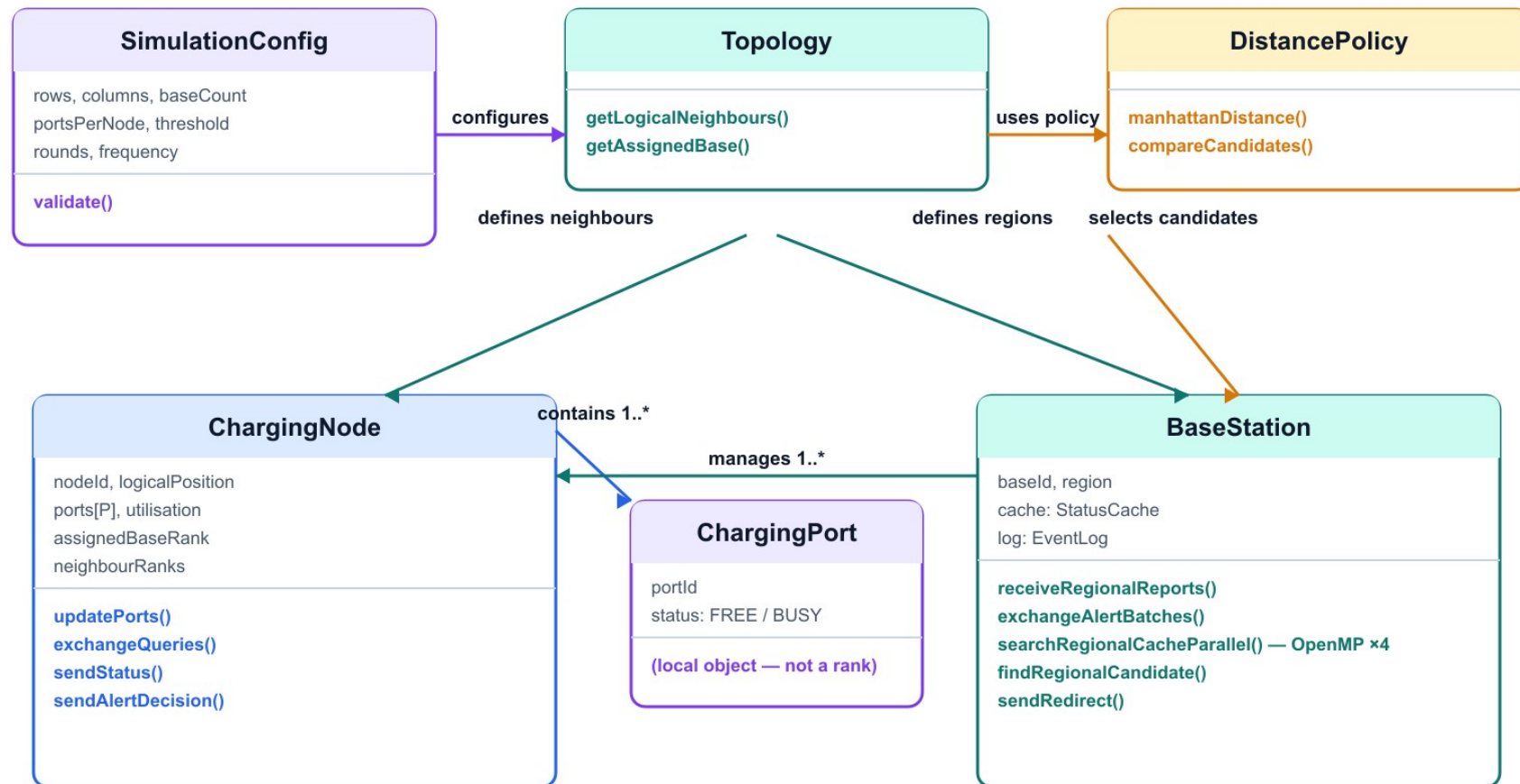
- The threshold, number of rounds, mesh dimensions and update frequency are user-configurable.
- Logical adjacency is independent of physical rank placement. The regular mesh is a reproducible simulation baseline.
- Charging ports remain local objects inside a charging-node process and do not create extra MPI ranks.

Static software structure

The diagram identifies every required simulation component and separates distributed processes from local state.

Task 2 · Static software structure (Hybrid MPI/OpenMP)

MPI processes model stations; ports remain local objects. Only Applied Week 5 APIs + MPI_THREAD_FUNNELED.



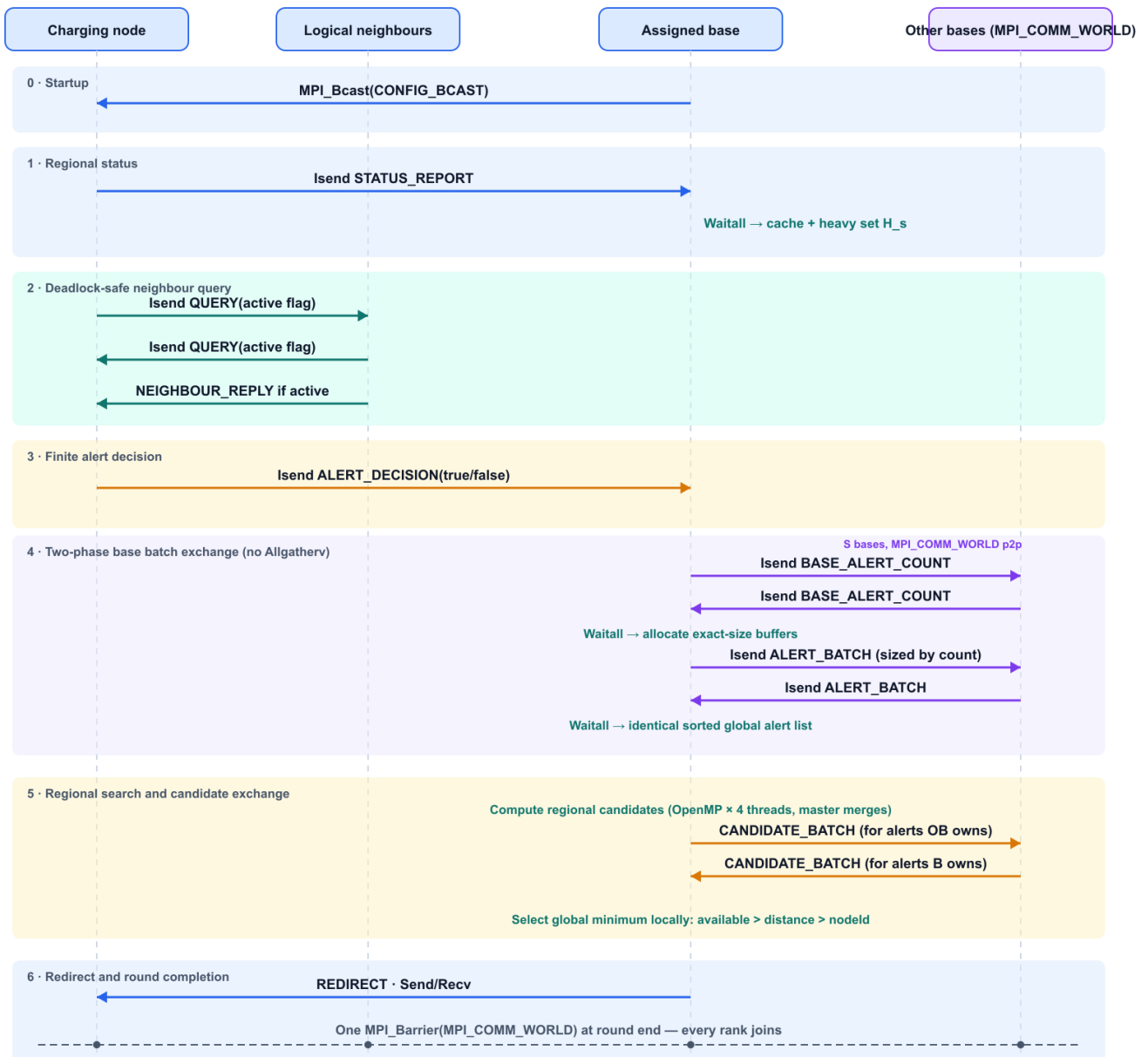
- ChargingNode and BaseStation are MPI processes. ChargingPort remains local state inside ChargingNode.
- Four OpenMP threads search disjoint cache partitions; the master thread merges their results.
- DistancePolicy can replace logical Manhattan distance with road distance or travel time.

Communication sequence for one round

Every phase has a known completion rule. Fixed query-intent records and count-before-batch exchange prevent indefinite receives.

Task 2 · Communication sequence for one round (Hybrid MPI/OpenMP)

Fixed completion rules for every phase; two-phase count-then-batch exchange replaces Allgather.



Only active query records request neighbour data. ALERT_DECISION(true) is the specification-level alert; false records complete the phase without creating an alert. OpenMP performs local cache search before the master thread exchanges candidate batches.

Rank layout and cluster deployment

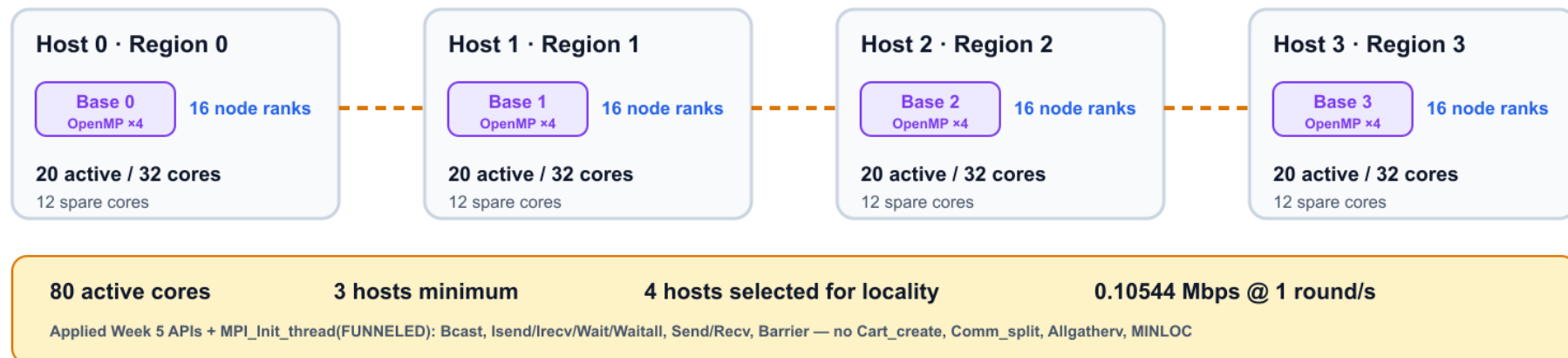
Ranks share MPI_COMM_WORLD. Rank arithmetic creates the logical mesh; physical placement groups each region with its assigned base.

Task 2 · Rank layout and cluster deployment (Hybrid MPI/OpenMP)

One flat MPI_COMM_WORLD; logical neighbours from rank arithmetic, not MPI_Cart_create.



Selected physical mapping · four 32-core hosts



- For nodeIndex = rank - S: row = nodeIndex / C and column = nodeIndex % C. Boundary checks return up to four valid neighbours.
- Each OpenMP team contains four total threads, including the master thread. MPI is initialised with MPI_THREAD_FUNNELED.

Message catalogue and sizing

Payload sizes are analytical assumptions and exclude MPI and lower-level network headers. Task 3 uses the same catalogue.

Message	Direction	Payload	MPI operation	Per-round scaling
CONFIG_BCAST	base 0 to all	64 B	Bcast	startup only
STATUS_REPORT	node to assigned base	32 B	Isend/Irecv/Waitall	N
QUERY	node to neighbour	16 B	Isend/Irecv/Waitall	2E_mesh
NEIGHBOUR_REPLY	neighbour to active node	24 B	Isend/Irecv/Waitall	up to 2E_mesh
ALERT_DECISION	heavy node to base	32 B	Isend/Irecv/Waitall	H; A true
BASE_ALERT_COUNT	each base to other bases	4 B	Isend/Irecv/Waitall	S(S-1)
ALERT_BATCH	each base to other bases	32 B/alert	Isend/Irecv/Waitall	A(S-1)
CANDIDATE_BATCH	base to alert owner	16 B/item	Isend/Irecv/Waitall	A(S-1)
REDIRECT	owner base to source node	24 B	Send/Recv	up to A
ROUND_SYNC	all ranks	0 B	Barrier	once

Correctness and bandwidth

Receives are posted before matching sends, non-blocking requests complete before buffer reuse, distinct tags separate message classes, and round identifiers reject stale records. OpenMP workers access local memory only; the master thread performs MPI calls under MPI_THREAD_FUNNELED.

For the four-host worst case with $A = N = 64$, external payload is 10,544 bytes per round: 1,280 bytes of cross-region neighbour traffic, 48 bytes of alert counts, 6,144 bytes of alert batches and 3,072 bytes of candidate batches. At one round per second with a 25 percent allowance, required aggregate bandwidth is $1.25 \times 8 \times 10,544 = 0.10544$ Mbps.

Limitations and future work. The baseline assumes reliable MPI ranks and uses logical Manhattan distance. Process-failure handling and a real road or travel-time distance source remain future extensions.